

Trust Core V1 Project

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1. Purpose & Value

What does it do?

This chip acts as a gatekeeper for the firmware running on defense and naval equipment. Before a device's software is allowed to start, the chip checks that the firmware has not been secretly modified. If tampering is detected, the device is blocked from booting.

Objective

PoC of hardware Root of Trust for ships

- Verifies firmware integrity
- Authenticates device identity
- Authenticates transmitted data

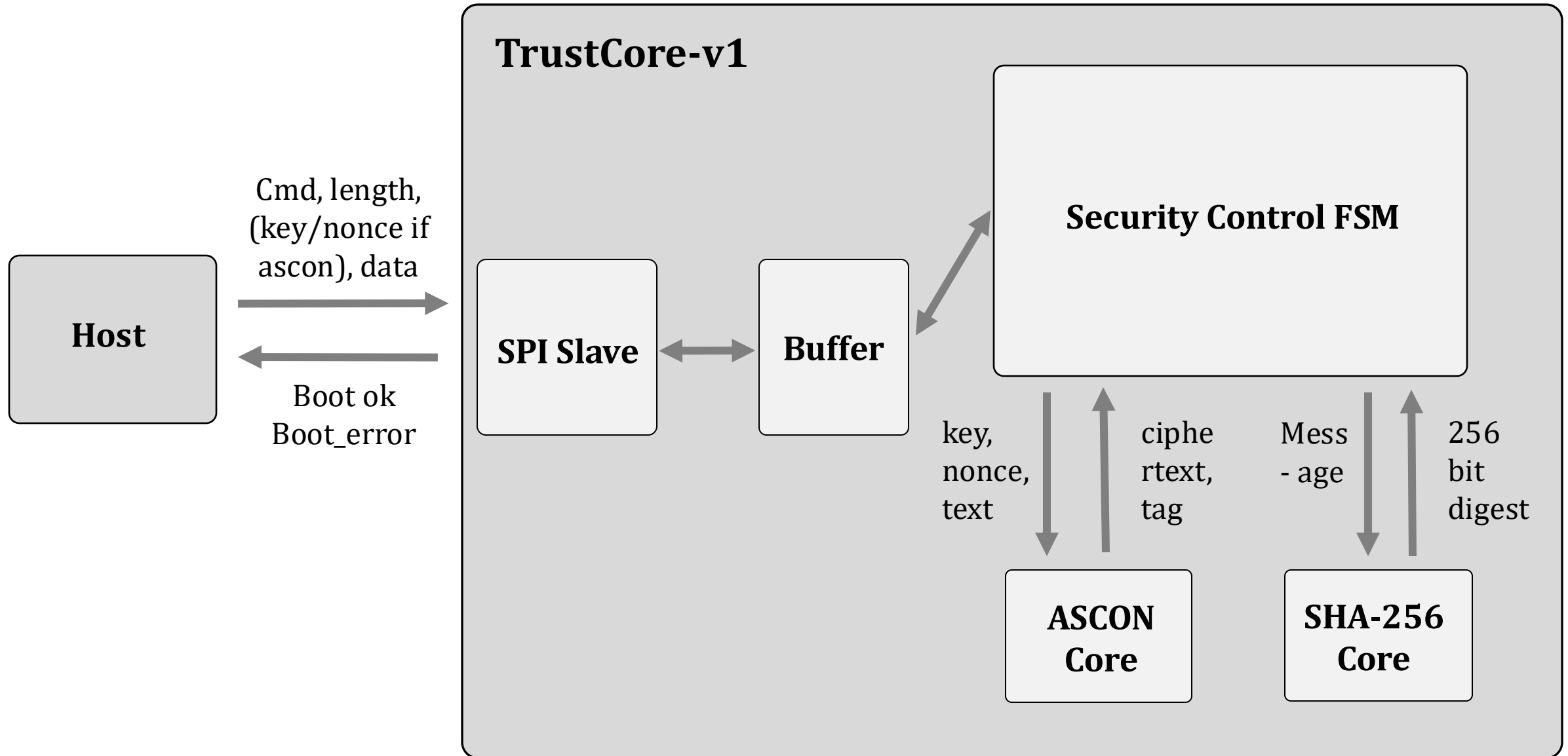
Where is it used?

- Shipboard sensor and control modules on naval vessels

Protects:

- | | |
|------------------------------|-------------------------------|
| • ECDIS | • VDR / Data Logger |
| • AIS / GNSS / Radar Gateway | • Shipboard Firewall/ Gateway |
| • PLC | • Autonomous Nav. Computer |
| • Propulsion Controller | • Low-power IoT sensors |
| • Steering Controller | |

2. Hardware Architecture



3. Project Specification — I/O & Timing Budget

I/O Definition

Pin	Name	Description
1	VDD	Supply voltage
2	VSS	Ground
3	CLK	System clock input
4	RST_N	Active-low reset
5	SCLK	SPI clock (from host)
6	MOSI	SPI data in (host → chip)
7	MISO	SPI data out (chip → host)
8	CS_N	SPI chip select (active low)

Target Clock Frequency

- System / core clock: 25 MHz
- SPI clock: 5 MHz (slower than the system clock)

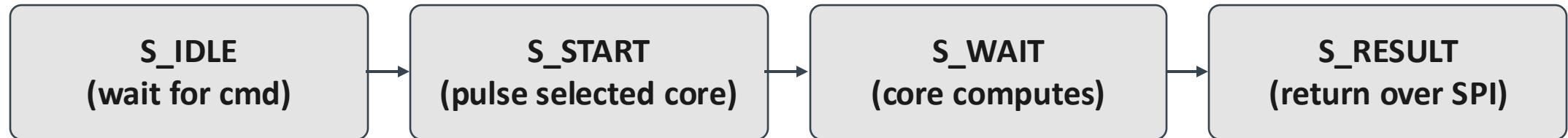
Estimated Gate Count (rough, pre-synthesis)

Block	Rough estimate
SHA-256 core	~10,000–15,000 GE
Ascon-128 core	~3,000–5,000 GE
SPI + buffer + crypto_fsm router	~1,000–2,000 GE
Security Control FSM (compare/boot-gate)	~350 GE
Total (rough)	~14,000–22,000 GE

4. Project Specification — Design Operations (SHA-256)

SHA-256 (what it is)

SHA-256 is a cryptographic hash function: it takes any input message and produces a fixed 256-bit (32-byte) digest. It is one-way (the digest can't be reversed back to the message) and highly sensitive to input changes -- flipping even one bit of the input changes the digest completely. That makes it well suited to detect tampering: any modification to firmware changes its digest.



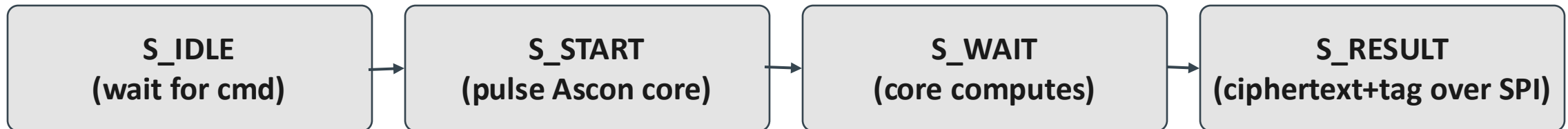
How a firmware integrity check works today

- Host sends the firmware bytes over SPI with CMD = SHA-256
- crypto_fsm routes the request to the on-chip SHA-256 core, waits for it to finish, and returns the digest to the host
- The host compares that digest against a known-good reference and decides whether to allow boot -- this decision is not yet made on-chip

4. Project Specification — Design Operations (Ascon-128)

Ascon-128 (what it is)

Ascon-128 is a lightweight authenticated encryption (AEAD) algorithm: a 128-bit key and 128-bit nonce drive a 320-bit internal state through a permutation, encrypting plaintext while also producing a 128-bit authentication tag. The tag lets a receiver confirm both that the data wasn't altered and that it came from someone holding the correct key. That's how this chip authenticates device identity and transmitted data.



How device / data authentication works

- Host sends key (16B), nonce (16B), and raw plaintext + length over SPI with CMD = Ascon
- spi_buffer_ctrl pads the plaintext internally to Ascon's required block size -- the host does not need to pre-pad
- crypto_fsm routes the request to the on-chip Ascon-128 core and waits for it to finish
- Core returns ciphertext (first 16 bytes) + 128-bit tag over SPI; the tag is what the host checks to authenticate the device and its data

5. Next Development Steps

